STATS 200A Lecture 6

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Definition 6.1 (Simple Function)

A simple function is measurable and takes finitely many values: $s = \sum_{i=1}^{k} a_i \mathbf{1}_{A_i}$, with disjoint level sets $A_i = \{s = a_i\}$.

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R})), \quad X = \sum_{i=1}^{k} a_i \mathbf{1}_{A_i}$$

$$\mathbb{E}(X) = \int X \, d\mathbb{P} = \sum_{i=1}^{k} a_i \mathbb{P}(A_i)$$

f, g are simply

$$\int (f+g)\, d\mu = \int f d\mu + \int g \, d\mu.$$

$$s_1^{(\omega)} \le s_2^{(\omega)} = \int s_1 \, d\mu \le \int s_2 \, d\mu$$

Lemma 6.2 (Approximation)

Every measurable $f \ge 0$ is a pointwise limit of nonnegative simple functions $s_n \uparrow f \quad (s_n \le s_{n+1}) \quad s_n \xrightarrow{p} f$.

$s_n = 2^{-n} \left[2^n (\min(f, n))\right]$

Theorem 6.3 (Monotone Convergence Theorem)

For measurable $0 \le f_n \uparrow f$ pointwise,

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

Theorem 6.4 (Fatou)

For measurable $f_n \ge 0$,

$$\int \liminf_n f_n \le \liminf_n \int f_n.$$

Theorem 6.5 (Dominated Convergence Theorem)

If measurable $f_n \xrightarrow{p} f$, and $|f_n| \leq g \in L^1$, then $f \in L^1$ and $\int |f_n - f| \to 0$, hence $\int f_n \to \int f$.

Theorem 6.6 (Change of Variables)

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (E, \mathcal{E})$ have law $\mathbb{P}_X$, and let $g : E \to \mathbb{R}$ be measurable. For $g \geq 0$ or $g \in L^1(\mathbb{P}_X)$

$$\mathbb{E}\left[g(X)\right] = \int_\Omega g(X(\omega))\mathbb{P}(d\omega) = \int_\mathcal{E} g(x)\mathbb{P}_X(dx).$$

Moreover, $g \in L^1(\mathbb{P}_x) \Leftrightarrow g(X) \in L^1(\mathbb{P})$.

$$(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (E, \mathcal{E}) \xrightarrow{g} (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$
$$\underset{g \circ X}{\curvearrowright}$$